

Rohit Kumar Debnath

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EDUCATION

Heritage Institute Of Technology July 2024– PRESENT
B.Tech in Computer Science and Engineering - YGPA - 9.35 Kolkata,WB, India

EXPERIENCE

Hive Dynamics March 2026 – PRESENT
SDE Intern Remote

- **Engineered Multi-Camera Video Analytics:** Refined YOLO and OpenCV pipelines for precise body detection and human activity tracking across active workspaces, enabling continuous automated productivity analysis.
- **Implemented Robust Person ReID & Facial Recognition:** Integrated ArcFace alongside deep Person Re-Identification (ReID) models to maintain consistent identity tracking of personnel across disparate camera angles and occlusions (temporary blockages)
- **Optimized High-Throughput MLOps Infrastructure:** Architected a containerized Docker pipeline utilizing HLS streaming protocols and Vector Databases to ingest live camera feeds, process high-dimensional embeddings, and deliver sub-millisecond similarity matching.
- **Integrated Multimodal AI Models:** Implemented Vision-Language Models (VLMs) to provide semantic context to the tracked behaviors, translating raw OpenCV/YOLO coordinates into high-level, actionable workflow insights.

PROJECTS

AlgoSlice | React JS | Python | Rust | CUDA | Docker March 2026

A prototype project on Algorand that explores remote access to GPU resources for developers who need computing power for heavy workloads. The application experiments with enabling users to utilize shared GPU infrastructure to run tasks without owning expensive hardware.

NanoBonds | ReactJS | NodeJS | ExpressJS | MongoDB | Solidity January 2026

NanoBonds is a decentralized finance (DeFi) platform that democratizes access to government bonds. It enables retail users to invest in fractional units of government bonds using stablecoins, providing a transparent, secure, and highly accessible investment experience through blockchain technology

TECHNICAL SKILLS

Languages: Python, C++, Java, JavaScript, Solidity
Technologies/Frameworks: ReactJS, NextJS, Open_CV, Docker, Nodejs, ExpressJS, Hardhat
Databases: Vector DB, MongoDB, PostgreSQL
Developer Tools: Git and Github, VSCode, Pycharm, Vercel, Netlify

ACHIEVEMENTS

Double Winner at Diversion 2026 Hackathon Finalist at IIT KGP EIBS 2.0 '26
Winner of Sourcify AI ML Track Winner
Official Contributor at Hacktoberfest '25